

WCET Analysis Methods

Milestone 3: Critical Evaluation and Transfer

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1 Goal of Milestone 3

- Evaluate the WCET method families beyond simple summary.
- Identify strengths, risks, and restrictive assumptions.
- Decide where each method is useful and where it can fail.
- Transfer the discussion to realistic embedded systems and safety-critical contexts.

2 Strengths of the Methods

Method	Main strength	Where this strength helps
SDTA	Can provide a mathematically sound deterministic upper bound.	Simple processors, safety-critical software, well-documented microcontrollers.
MBDTA	Uses real execution data from the target hardware.	Industrial platforms where measurement is easier than modeling.
HYDTA	Improves pure measurement by adding static path information.	Programs where testing alone gives weak path coverage.
SPTA	Gives a probabilistic argument from static reasoning.	Research settings with simple randomized hardware models.
MBPTA	Can produce probabilistic WCET estimates for complex hardware.	Promising direction for multicore and mixed-criticality systems.
HYPTA	Combines probabilistic reasoning with reduced testing effort.	Useful when full measurement campaigns are too expensive.

3 Main Limitations and Risks

- **General multicore risk**
 - Most WCET methods were originally designed around single-core assumptions.
 - On multicore processors, tasks interfere through shared caches, buses, and memory controllers.
 - A fixed safety margin over measured time can become unsafe when interference changes strongly.
- **SDTA risks**
 - Requires a complete and correct hardware timing model.
 - Vendor documentation may not describe all timing-relevant behavior.

- Manual flow facts, such as loop bounds, can be wrong.
- For complex processors, the analysis may become too expensive or too pessimistic.
- **MBDTA risks**
 - The real worst-case path may never be executed during testing.
 - Test inputs may not cover rare hardware states.
 - Adding a simple fixed margin, for example 20%, is easy but not scientifically reliable.
- **Hybrid risks**
 - Path coverage improves, but hardware-state coverage is still difficult.
 - Separately measured code segments may not reproduce the real cache and memory state.
- **Probabilistic risks**
 - Statistical assumptions must hold, otherwise the pWCET result can be misleading.
 - MBPTA often needs hardware time randomization that is not available in normal production processors.
 - Certification arguments are still weaker than for established deterministic methods.

4 Restrictive Assumptions

Method	Assumption	When it breaks down
SDTA	Hardware model is correct and complete.	Processor has undocumented behavior, errata, caches, or complex pipelines.
SDTA	Program flow facts are known.	Legacy code, data-dependent loops, recursion, or missing annotations.
MBDTA	Test inputs cover the worst-case behavior.	Large input space or rare combinations of data and hardware state.
HYDTA	Measured blocks can be safely combined.	Cache, branch predictor, or memory state changes between separate runs.
MBPTA	Measurements are representative and statistically valid.	Randomization is missing, disabled, or not strong enough.
All methods	Test/development platform matches deployment platform.	Compiler changes, OS changes, hardware revisions, or different clock/memory settings.

5 Scalability and Implementation Implications

Aspect	Observation	Engineering implication
Program size	More paths and loops increase analysis effort.	Manual annotations and test design become harder.
Hardware complexity	Caches, pipelines, and predictors increase timing-state space.	Static models become expensive and difficult to validate.

Multicore execution	Shared resources cause interference between tasks.	Analysis may require isolation, worst-case contention modeling, or new methods.
Measurement campaigns	More tests improve confidence but cost time and storage.	Practical projects must balance test effort with evidence quality.
Certification	Not every technically good method is accepted by standards.	Engineers need arguments that auditors can understand and trust.
Tool availability	Industrial tools are stronger for classical single-core analysis.	Newer methods may remain academic even if they solve important problems.

6 Suitable and Unsuitable Application Examples

Method	Suitable example	Unsuitable example
SDTA	Engine control unit on a simple in-order microcontroller.	Modern application processor with out-of-order execution and shared cache.
MBDTA	Mature single-core railway system with stable platform and strong test history.	Autonomous driving software with huge input space and changing environment.
HYDTA	Embedded control code where path structure is known but testing is incomplete.	System with strong cache interaction across many code segments.
MBPTA	Research prototype for mixed-criticality multicore timing analysis.	Certified product today without hardware randomization support.
SPTA / HYPTA	Academic evaluation of probabilistic timing assumptions.	Direct industrial certification without mature tool and hardware support.

7 Possible Extensions and Open Research Questions

- **Multicore WCET:** find practical ways to bound interference under realistic workloads.
- **Resource isolation:** combine WCET methods with cache partitioning, memory throttling, or time-triggered access to shared resources.
- **Automatic flow facts:** reduce manual loop-bound and path annotations in SDTA.
- **MBPTA hardware support:** design processors that support timing randomization without large performance loss.
- **Certification of probabilistic results:** define clear evidence rules for pWCET values.
- **AI and neural network timing:** analyze timing of embedded inference workloads with large memory access and accelerator use.
- **Online adaptation:** study whether runtime measurements can safely update WCET evidence after deployment.

8 Current Critical Judgement

- There is no universally best WCET method.
- The important question is: which assumptions are true for the target system?
- SDTA is strongest for simple and well-documented hardware.
- MBDTA is practical but risky if input coverage is weak.
- MBPTA is the most interesting direction for complex and multicore systems, but it still needs hardware and certification support.
- The biggest open gap is modern high-performance embedded hardware, especially multicore systems used in autonomous and AI-enabled applications.

References

- [1] R. Wilhelm et al., “The worst-case execution time problem: Overview of methods and survey of tools,” *ACM Transactions on Embedded Computing Systems*, vol. 7, no. 3, pp. 1-53, 2008.
- [2] L. Cucu-Grosjean et al., “Measurement-based probabilistic timing analysis for multi-path programs,” in *Proc. Euromicro Conference on Real-Time Systems (ECRTS)*, 2012.